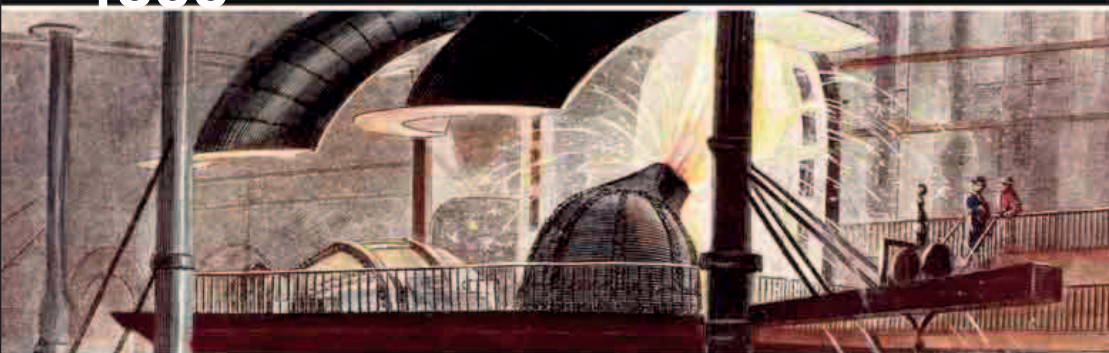
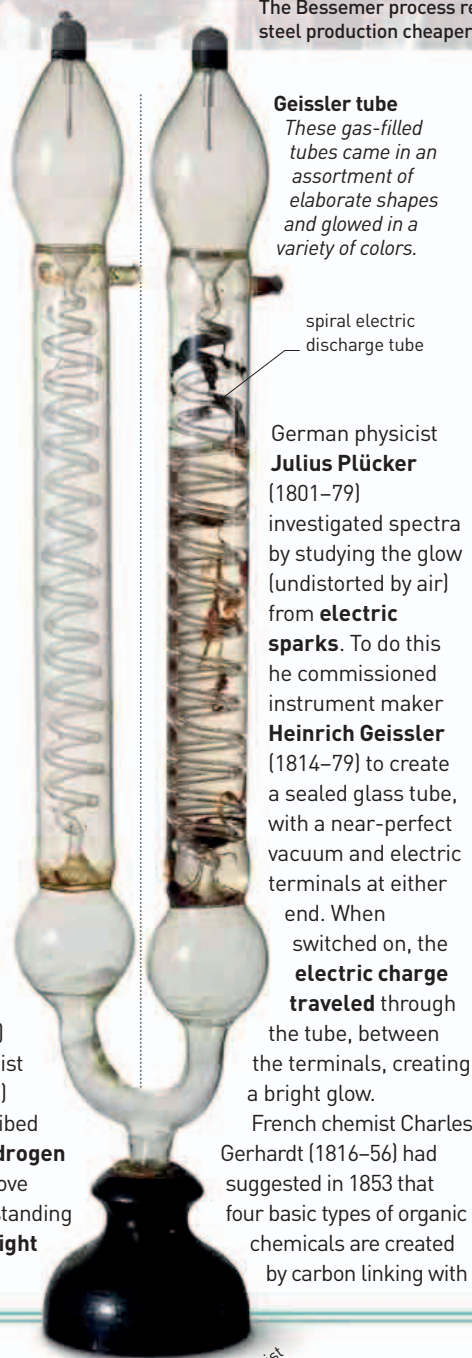




The Bessemer process revolutionized engineering by making steel production cheaper and more efficient.

AT THE FOREFRONT OF SCIENCE

In 1855 was the search for the relationship between **atoms, light, and electromagnetism**. British mathematician **James Clerk Maxwell** (1831–79) began working on a theory to unify electricity, light, and magnetism, while other scientists conducted practical experiments to find out how **atoms emitted light**. They had already worked out that each kind of atom emits and absorbs a particular range of colors, or spectrum, with dark lines (gaps) at some wavelengths and bright lines (peaks) at others. Swedish physicist **Anders Ångström** (1814–74) and American scientist **David Alter** (1807–81) independently described the **spectrum of hydrogen gas**, which would prove crucial in the understanding of the link between **light and atoms**.



Geissler tube
These gas-filled tubes came in an assortment of elaborate shapes and glowed in a variety of colors.

spiral electric discharge tube

German physicist **Julius Plücker** (1801–79) investigated spectra by studying the glow (undistorted by air) from **electric sparks**. To do this he commissioned instrument maker **Heinrich Geissler** (1814–79) to create a sealed glass tube, with a near-perfect vacuum and electric terminals at either end. When switched on, the **electric charge traveled** through the tube, between the terminals, creating a bright glow. French chemist **Charles Gerhardt** (1816–56) had suggested in 1853 that four basic types of organic chemicals are created by carbon linking with

hydrogen, hydrogen chloride, water, or ammonia molecules. In 1855, English chemist **William Odling** (1829–1921) added a fifth type, based on methane. This led German chemist **Friedrich Kekulé** (1829–96) and Scottish chemist **Archibald Couper** (1831–1892) to begin developing a **structural theory of molecules**.

“ATOMS WERE GAMBOLLING BEFORE MY EYES...”

Frederich Kekulé, German chemist, describing a daydream that led to his structural theory, 1855

The technological breakthrough of the year was the development of a special furnace, developed by the British engineer **Henry Bessemer** (1813–98), which allowed **steel to be made cheaply** and in quantity from pig iron (a crude form of iron).

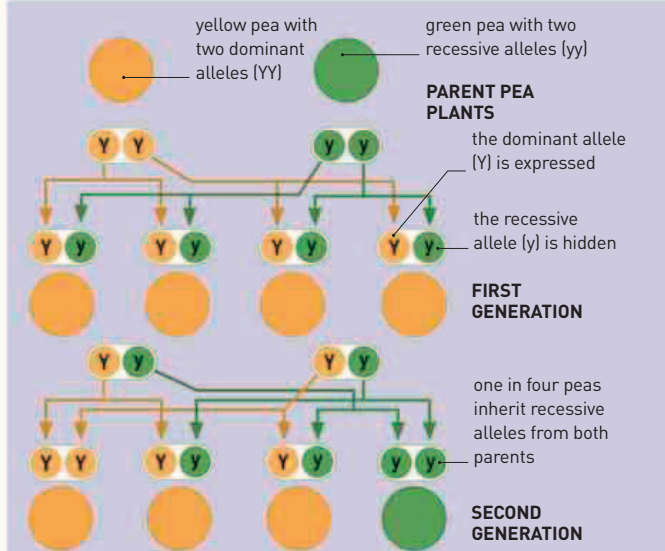
In Germany, an unusual fossil was found at Riedenburg. It was thought to be a flying reptile until 1970 when it was finally shown to have feathers, identifying it as the **first Archaeopteryx** ever discovered. It is evidence that birds evolved from the dinosaurs.



Montgomerie surveyed the Karakoram range as part of the Great Trigonometric Survey of India. It includes K2, the second-highest mountain in the world.

THE SCIENCE OF GENETICS BEGAN WITH THE WORK of the Austrian monk **Gregor Mendel** (1822–84), but his work was not fully appreciated until years later. Mendel began to experiment with pea plants in his monastery garden in 1856. He laid the foundations of genetics as he showed how **characteristics are**

passed on from generation to generation through what he called “factors,” which later became known as genes, that are inherited from both parents. Another discovery that was not fully appreciated until much later, was the discovery of the first recognized **fossil of a human ancestor**. In August



DOMINANT AND RECESSIVE GENES

Mendel's work on peas showed that inherited characteristics, such as color, are determined by particles, later called genes. Genes come in different forms, called alleles, that combine in different ways in offspring. A dominant Y allele determines yellow pea color and a single one of these being present is sufficient to make a yellow pea. A recessive y allele determines green pea color and two of these must come together to make a green pea.

Swedish physicist **Anders Ångström** publishes **spectrum of hydrogen**

British chemist **William Perkin** discovers **aniline dye mauveine**

American scientist **David Alter** describes the spectra of hydrogen and other gases

German chemist **Friedrich Kekulé** starts to develop the **structural theory of organic molecules**

May 10 German chemist **Robert Bunsen** invents the **Bunsen burner**

British mathematician **James Clerk Maxwell** begins work on his theory to unite light, electricity, and magnetism

October 17 Henry **Bessemer** patents the **Bessemer process of steelmaking**

The first **Archaeopteryx** fossil was discovered at **Riedenburg** in Germany

Austrian monk **Gregor Mendel** starts his research on plant inheritance

American meteorologist **William Ferrel** explains wind circulation

August German teacher **John Fuhlrott** identifies fossil bones of **Neanderthal man**

British surveyor **Thomas Montgomerie** names the **Himalayan mountain peaks K1 and K2**